Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/15/2025
PI Name:	Schubert, Michael Ph.D.		
Project Title:	Ground Validation of Self-Administered Incremental Rehabilitation Tool to Mitigate Motion Sickness and Enhance Sensorimotor Recovery		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21287-6921	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	01/01/2021	**End Date:**	09/30/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	4
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	End date changed to 09/30/2026 per the Principal Investigator (Ed., 10/6/25) End date changed to 09/30/2025 per NSSC information (Ed., 12/1/22)		
Key Personnel Changes/Previous PI:	October 2025 Update: Per the Principal Investigator (PI), Preston Adam, Ph.D. is no longer with the investigation (Ed., 10/23/25). Olivia Slomski has been removed as a co-investigator from the NAMRU-D site. No other changes have been made to the Key Personnel.		
COI Name (Institution):	Wood, Scott Ph.D. (NASA Johnson Space Center) Migliaccio, Americo Ph.D. (Neuroscience Research Australia) Folga, Richard (Naval Medical Research Unit in Dayton OH) Grove, Colin R Ph.D. (Emory University) Stewart, Matthew M.D., Ph.D. (Johns Hopkins University) Ludwig, Daniel (Johns Hopkins Hospital)		
Grant/Contract No.:	80NSSC21M0057		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Astronauts returning from long duration spaceflight suffer from motion sickness, vertigo, and postural imbalance that risk their safety during and after landing. Vestibular patients typically suffer from similar problems that risk their safety during activities of daily living. For both groups, rehabilitation using head motion is the key to recovering from these symptoms, but current methods are uncontrolled and non-quantified. Our team has successfully implemented a self-administered rehabilitation protocol that can be performed by patients at home to improve vestibular function. Our current system measures head and eye movements to improve vestibulo-ocular reflexes. We propose to modify our system to provide additional feedback on head motion to reduce motion sickness for both astronauts and patients as they undergo rehabilitation. We will compare motion sickness and recovery following +3Gx centrifugation (Spaceflight analog, Objective I) in two groups: a treatment group given feedback to guide their head motion and a control group with no specific head movement strategy. We will also perform similar measurements in patients recovering from acute vestibular loss (Earth analog, Objective II). We hypothesize this approach will result in a greater ability to tolerate head movements with fewer motion sickness symptoms. In addition to mitigating motion sickness and improving recovery when returning to Earth, our self-administered approach will enable astronauts to be more autonomous without the aid of their reconditioning experts during exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project innovates upon two patented technologies developed from principal investigator (PI) Michael Schubert and co-investigator Americo Migliaccio (US20100198104 and US20160242642A1, https://), by refining users' ability to self-treat motion sickness. The device we have built for use in this project guides users to perform active head rotations, matched to a metronome, about the yaw, pitch, and roll axes (90 sec epochs, 5 minutes per axis, 15 min total). In addition, the device includes the capacity for a subject to rate his/her/their perception of motion sickness using a handheld controller and integrates a heart-rate monitor worn over the subject's arm. Video-oculography captures eye and head velocity, and also tracks the number of blinks and heart rate variability – metrics that can indicate worsening nausea. The benefits of this research to life are similar and critical in both space and Earth environs – validation of an autonomous treatment for motion sickness and balance disorders.
Task Progress:	Objective I. We are using the 711TH HUMAN PERFORMANCE WING HUMAN-RATED CENTRIFUGE device located within the Air Force Research Lab, which required major edits to the human protection protocol. The IRB was approved on August 28, 2025. We have conducted three "dry-runs" with human subjects using the 711th HPWc, exposing subjects to 60 minutes of rotation at 3Gx. The dry-runs have been critical to ensure the safety and comfort of the subjects as well as ensuring the 711th HPWc motion profile is effectively generating motion sickness similar to what crewmembers experience on long-duration space travel. We were able to confirm an adequate intensity of motion sickness in all subjects. Objective II. We continue to obtain the names of surgical candidates at JHU as possible recruits. To date, we have collected data from n=16 (n=24 planned) patients with unilateral vestibular hypofunction due to having their balance nerve (vestibular nerve) cut to remove the benign vestibular schwannoma. A 17th subject was recruited but failed to show up for data collection. Each patient has been measured at 3 separate time points (pre-operative, inpatient stay day 2-5, 14-day follow-up). To date, we have analyzed outcome measures (including inertial measurement units /IMU data) from n=14 subjects measured at JHU. Due to complications during recovery, some data are missing. Below are results of our Oculomotor and Rehabilitation outcomes (Gait/Balance). Oculomotor: The SWAN device correctly identifies abnormal vestibulo-ocular reflex from the video head impulse test. Our measure of upright perception (vrVANTAN) includes two unique measures of perception of ocular alignment (vertical and torsion). The task is to align two colored lines so they appear as a single line in both vertical and torsional planes. • Our data show that ocular alignment significantly worsens post-operatively and does not return to Pre-Operative values when the final measure is recorded at post-operative day 14. • The asymmetries of vertical misalignment seem to improve (Figure 5), while the asymmetry in torsional alignment persists. Rehabilitation: Eight subjects have been assigned to SWAN, while n=6 subjects were assigned to the control group of traditional vestibular rehabilitation. We are comparing IMU data between the groups during four balance tasks: gait velocity (10M walk test), gait endurance (2 min walk test), Timed Up and Go (TUG), and standing balance (stand on foam, eyes open/closed, head neutral/tilted up). • The root mean square (RMS) of head velocity between the two groups (SWAN vs Control) was no different at POD 14 for either the 10M walk (range 1.5 –3.7) or the 2 Minute Walk tests (range 1.3 –3.1). • During the TUG test, the Walk:Turn ratio of time was 1.5 times slower in the control group at a clinically meaningful difference, yet not statistically significant – likely reflecting our current sample size for each cohort. • SWAN participants' head sway was always smaller than the control group by 1.5x, standing on foam eyes open or closed. • All subjects have a greater sway path of the head when standing on foam, eyes closed, head tilted up. In summary, our early analysis of this incomplete sample suggests that perception of alignment remains abnormal at least 2 weeks post-operatively. In addition, both static and dynamic measures of postural control are improved after the SWAN rehab (TUG and CTSIB) compared to traditional vestibular rehabilitation.
Bibliography Type:	Description: (Last Updated: 10/23/2025)
Conference Materials (Downloadable)	Schubert MC, Ludwig D, Wood SJ, Grove CR, Folga RV, Stewart CM. "Self-administered rehabilitation tool to mitigate motion sickness and enhance sensorimotor recovery." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 SWAN IWS 2025.pdf